

ARTICLE TYPE

Comparing Linear Dynamical Systems via VAR approximation and Isotropic Distance

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Abstract

We study the problem of determining whether two observed time series are generated by the same underlying linear dynamical system, when latent parameters are inaccessible. Each observed process is represented via a finite-order VAR approximation, and systems are compared through a distance defined in the induced parameter space. This approach exploits the fact that differences in latent dynamics are faithfully reflected in the observable autoregressive structure. We formalize the comparison as a hypothesis test and evaluate a threshold-based decision rule through systematic Monte Carlo experiments over model order, system dimensionality, sample size, dynamical regime separation, and observation noise. The method achieves near-perfect classification under moderate conditions, degrades gracefully with increasing model complexity or reduced sample size, and is notably robust to observation noise.

Keywords: Linear Dynamical Systems (LDS), Vector Autoregression (VAR), Least Squares Estimation (LSE), Time Series Analysis, Systems Identification

1. Introduction

Often in classical settings, dynamical systems arise from first principles. Take, for example, Newton's 2nd law of motion (i.e., $F = ma$), which provides a direct mechanism for constructing the underlying system matrices governing the evolution of a system. In cases such as these, the dynamics are specified a priori, and the resulting trajectories are obtained by *forward* simulation.

In stark contrast, the setting considered in this work is fundamentally *inverse*. The underlying system parameters are not accessible, and only observations of processes are available to us. Consequently, one must work *backwards* from observed time series to infer a representation of the underlying LDS dynamics. Our problem therefore lies within the broader context of system identification, where the goal is to recover meaningful structure from data rather than derive it from known physical laws, as in the case of populating system matrices via Newton's 2nd law.

This inverse perspective arises in a myriad of real-world settings. For example:

- **Economics:** where relationships between macroeconomic indicators must be inferred from observations in time series.
- **Neuroscience:** where neural dynamics are reconstructed from EEG or fMRI signals.
- **Astrophysics:** where one observes stellar light curves or spectral measurements as time series and must infer the underlying dynamical process (e.g., stellar oscillations, accretion dynamics, orbital motion).

In this present work, we address the problem of comparing dynamical systems under such observational constraints. Rather than attempting to recover latent parameters directly,

we instead approximate each system through its induced autoregressive representation and compare systems in this observational parameter space.

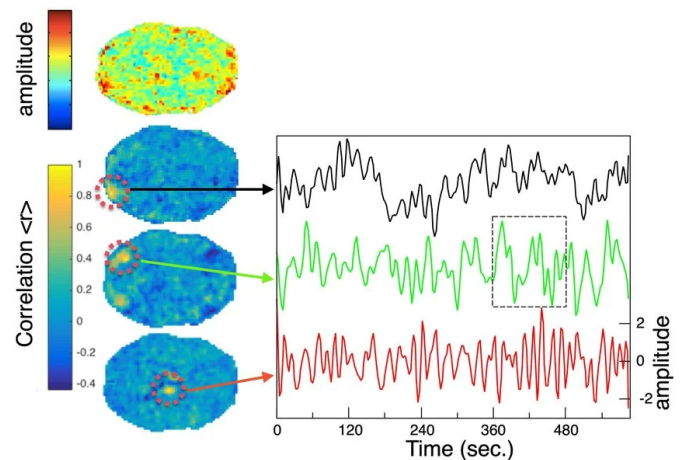


Figure 1. Time series data derived from EEG signals (see Cifre et al. 2020)

More precisely, given two time series, we seek to determine whether they are generated from the same underlying dynamical system or by distinct systems. To address this, we map each observed process to a finite-dimensional autoregressive representation via $VAR(p)$ approximation, estimate its parameters using least squares, and compare systems through a distance defined in parameter space.

A principal characteristic of our approach is that it operates entirely in the observable domain, avoiding the need for explicit system identification of the latent dynamics. The effectiveness of this representation is governed by the stability of the induced dynamics, which controls the extent to which the

observed process can be well-approximated by a finite-order autoregressive model.

2. Model and Problem Formulation

2.1 Linear Dynamical System

Consider a linear dynamical system (LDS) describing the evolution of a latent state observed through noisy measurements. Such systems are commonly represented in state-space form (see Tsiamis and Pappas 2021) as:

$$\begin{aligned} x_{k+1} &= Ax_k + L\epsilon_k \\ y_k &= Cx_k + \epsilon_k \end{aligned} \quad (1)$$

Where $x_k \in \mathbb{R}^{d_x}$ denotes the latent state, $y_k \in \mathbb{R}^{d_y}$ the observed time series, and $\epsilon_k \in \mathbb{R}^{d_y}$ a series of independent noise vectors. In the general state-space formulation, an additional control input term of the form Bu_k may be present, which represents exogenous inputs driving the system. For our considerations, however, we restrict our attention to the autonomous setting where no external inputs are present. This allows us to focus solely on the intrinsic dynamics of the system as reflected in the observed process.

This formulation captures a broad class of systems where the observed process y_k is driven by an underlying latent structure. As previously discussed, in numerous practical settings, the matrices of our system, i.e., (A, C, L) , are unknown, and only realizations of the observed process are readily available.

2.2 Problem Formulation

Given the observational setting described above, we now consider the task of comparing dynamical systems based solely on their observed trajectories.

Let $y^{(1)}$ and $y^{(2)}$ denote two observable time series, each generated by potentially distinct linear dynamical systems of the form given in (1). We then seek to determine whether $y^{(1)}$ and $y^{(2)}$ arise from the same underlying system or from different systems.

Formally, this can be expressed as a hypothesis testing problem:

$$H_0 : (A^{(1)}, C^{(1)}, L^{(1)}) = (A^{(2)}, C^{(2)}, L^{(2)}) \quad (\text{same system})$$

$$H_1 : (A^{(1)}, C^{(1)}, L^{(1)}) \neq (A^{(2)}, C^{(2)}, L^{(2)}) \quad (\text{different systems})$$

Since the latent parameters are not accessible, this decision must be made purely on the basis of the observed sequences. Direct comparison in the latent parameter space is therefore not feasible, and one must instead construct a representation of each of these observed processes that capture the underlying system dynamics in a comparable form. This begets the need for a representation that operates entirely in the observable domain while still adequately capturing the underlying system dynamics.

Thus, we take an approach based on autoregressive representations of the observed processes. Each time series is

mapped to a finite-dimensional parameter vector obtained from a $\text{VAR}(p)$ approximation, and comparisons between systems are performed via a distance defined in this induced parameter space.

Hence, the central question of this work is whether distances between these representations can reliably distinguish between realizations drawn from the same underlying system and those drawn from different underlying systems.

3. Methodology

In this section, we describe the proposed theoretical pipeline for comparing dynamical systems from observed time series. The approach consists of three main steps: (i) representing each observed process via a finite-order autoregressive model, (ii) estimating the corresponding parameters from data, and (iii) comparing systems through a distance defined in this resulting parameter space.

3.1 Autoregressive Representation

Although the system evolves in latent space (i.e., the unobserved state x_k), it can be expressed purely in terms of past observations. By recursively substituting the state equation into the observation equation, one obtains the following *exact* finite-order representation (whose proof may be found in Appendix 1):

Lemma 3.1

$$y_k = \underbrace{C(A - LC)^p x_{k-p}}_{\text{latent term}} + \underbrace{\sum_{i=1}^p C(A - LC)^{i-1} L y_{k-i}}_{\text{autoregressive component}} + \underbrace{\epsilon_k}_{\text{noise}} \quad (2)$$

$$y_k \approx \sum_{i=1}^p \Phi_i y_{k-i} + u_k, \quad \Phi_i = C(A - LC)^{i-1} L \in \mathbb{R}^{d_y \times d_y} \quad (3)$$

(2) decomposes the observed process into three distinct components: a vanishing latent term, an autoregressive component and a noise term. The latent term arises from its dependence on the unobserved state x_{k-p} , while the autoregressive component captures the recursive dependence of y_k on its past observations, i.e., y_{k-i} . The final term ϵ_k corresponds to the observation noise.

Since the latent state is not directly observable, this term captures information that is not encoded in the observed history of the process. Under the stability condition:

$$\rho(F) < 1, \quad \rho(F) = \max_i |\lambda_i(F)|$$

where

$$F := A - LC \quad (4)$$

the matrix powers F^p decay geometrically as p increases. Consequently, the influence of the latent state on the current observation diminishes over time, and for sufficiently large

p , this contribution becomes negligible. It is precisely this negligibility which allows us to approximate the observed process in (3) as depending only on past observations, leading to a finite-order autoregressive representation of the underlying system.

This autoregressive representation provides the foundation for our approach, because it allows us to encode the underlying system dynamics entirely in terms of observable quantities.

Given that we only have access to observations of the process, we now estimate the parameters of the VAR(p) representation and use them to compare two dynamical systems.

3.2 Regression Formulation

To estimate the autoregressive coefficients from data, we express the VAR(p) model as a linear regression problem (see Jonáš 2009). Given a realization γ_k , we construct the regression matrices as follows:

$$X = \begin{bmatrix} \gamma_p^\top & \gamma_{p-1}^\top & \cdots & \gamma_1^\top \\ \gamma_{p+1}^\top & \gamma_p^\top & \cdots & \gamma_2^\top \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1}^\top & \gamma_{n-2}^\top & \cdots & \gamma_{n-p}^\top \end{bmatrix}$$

$$Y = \begin{bmatrix} \gamma_{p+1}^\top \\ \gamma_{p+2}^\top \\ \vdots \\ \gamma_n^\top \end{bmatrix} \quad B = \begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \vdots \\ \Phi_p \end{bmatrix} \quad U = \begin{bmatrix} u_{p+1}^\top \\ u_{p+2}^\top \\ \vdots \\ u_n^\top \end{bmatrix}$$

These matrices satisfy the linear regression relationship:

$$Y = XB + U \quad (5)$$

where $B \in \mathbb{R}^{pd_\gamma \times d_\gamma}$ contains the VAR coefficient matrices $[\Phi_i]_{i=1}^p$ stacked vertically, and $U \in \mathbb{R}^{T \times d_\gamma}$ collects the innovation terms.

3.3 Least Squares Estimation

Given the aforementioned regression formulation (5), we estimate the coefficient matrix B via least squares (see Jonáš 2009):

$$\hat{B} = (X^\top X)^{-1}(X^\top Y).$$

The estimated matrix $\hat{B}$ can be decomposed into blocks which correspond to the autoregressive coefficients $(\hat{\Phi}_1, \dots, \hat{\Phi}_p)$. This estimator minimizes the Frobenius norm of the residual matrix, or more formally,

$$\hat{B} = \arg \min_B \|Y - XB\|_F^2.$$

whose solution is given in Appendix 2

This estimator provides a consistent estimate of the VAR coefficients under standard assumptions, effectively modeling how the observed process depends on its past.

3.4 Isotropic Distance Metric

Given two observed time series, we compute their corresponding parameter estimates $\hat{\Phi}^{(1)}$ and $\hat{\Phi}^{(2)}$. We define the isotropic distance between systems as:

$$D = \sum_{i=1}^p \left\| \hat{\Phi}_i^{(1)} - \hat{\Phi}_i^{(2)} \right\|_F^2, \quad (6)$$

which gives us a scalar measure of dissimilarity, quantifying the squared Euclidean separation between the two systems in the space of the autoregressive parameters (i.e., $\mathbb{R}^{pd_\gamma^2}$).

3.5 Decision Rule

Given the distance metric defined above, we formulate a threshold-based decision rule for the hypothesis testing problem introduced in Subsection 2.2. In particular, we classify pairs of time series according to

$$\text{Decide } \begin{cases} H_1 & \text{if } D > \tau, \\ H_0 & \text{if } D \leq \tau, \end{cases}$$

where τ is a threshold that separates distances corresponding to pairs generated from the same underlying system from those generated by distinct systems.

In practice, the distributions of D under H_0 and H_1 are estimated empirically via Monte Carlo simulation, and the choice of τ determines the trade-off between false positives and false negatives. To evaluate classification performance, we consider standard metrics including the true positive rate (TPR), false positive rate (FPR), receiver operating characteristic (ROC) curves, area under the curve (AUC), and F1 score.

This reduces the problem of comparing dynamical systems to a statistical decision problem in a one-dimensional space, where separation between the distributions of D under H_0 and H_1 determines the effectiveness of our method.

4. Experimental Design

We now describe the experimental framework used to evaluate the proposed comparison method. In particular, we outline the generation of synthetic linear dynamical systems and the simulation procedures used to assess separation between same-system and different-system pairs.

4.1 System Sampling

To evaluate the proposed comparison framework, we generate synthetic linear dynamical systems of the form given in (1), with latent dimension d_x and observation dimension d_γ .

A central aspect of our experimental design is the control of the effective memory of the system, which is governed by the matrix F given in (4). As shown in Subsection 3.1, the matrix F determines the decay of the autoregressive coefficients and therefore the extent to which the observed process can be approximated by a finite-order VAR model.

To directly control this behavior, we adopt a sampling procedure in which F is drawn first. Specifically, we sample a

random matrix with i.i.d. Gaussian entries and rescale it such that its spectral radius lies within a prescribed interval:

$$F_{\text{raw}} \sim \mathcal{N}(0, 1)^{d_x \times d_x}$$

$$F = \frac{\rho_{\text{target}}}{\rho(F_{\text{raw}}) + 10^{-12}} F_{\text{raw}}, \quad (7)$$

where

$$\rho_{\text{target}} \in [\rho_{\min}, \rho_{\max}]$$

Given F , we then construct the system matrix A via

$$A = F + LC,$$

which ensures that the relationship $F = A - LC$ holds exactly.

The observation matrix $C \in \mathbb{R}^{d_y \times d_x}$ is sampled from a standard Gaussian distribution, i.e., $C \sim \mathcal{N}(0, 1)$, while the noise injection matrix $L \in \mathbb{R}^{d_x \times d_y}$ is taken to be the identity. This choice ensures that noise enters the system isotropically, without introducing additional directional structure.

Additionally, alternative sampling strategies for (C, L) are employed when sweeping model parameters to ensure consistency in the induced system structure across different configurations (see Appendix 3).

4.2 Trajectory Simulation

Given a sampled system (A, C, L) , we simulate trajectories according to the linear dynamical system defined in (1). In particular, the latent state evolves as

$$x_{k+1} = Ax_k + L\epsilon_k,$$

and the observed process is given by

$$y_k = Cx_k + \epsilon_k,$$

where the noise sequence satisfies

$$\epsilon_k \sim \mathcal{N}(0, \sigma^2 I).$$

For each realization, we generate an observed trajectory $[y_k]_{k=1}^n$ of length n . While the latent states x_k are used in the simulation, only the observed sequence y_k is retained, reflecting the observational setting described in Section 2.2.

These observed trajectories serve as the input for $\text{VAR}(p)$ estimation in the subsequent analysis, from which the autoregressive parameters used for system comparison are obtained.

4.3 Monte Carlo Structure

Each experiment is organized as a two-level Monte Carlo procedure. In the outer loop, we sample a fixed collection of dynamical systems, referred to as realizations. In the inner loop, we repeatedly simulate observed trajectories from these fixed systems and fit a $\text{VAR}(p)$ model to each trajectory, referred to as trials. Crucially, this design allows us to separate variability arising from the underlying system dynamics from variability induced by observation noise and finite-sample estimation. Unless otherwise specified, all experiments are conducted using 25 realizations and 40 Monte Carlo trials per realization.

4.4 Construction of H_0 and H_1 Pairs

For each fixed realization, we construct two distinct classes of pairwise comparisons corresponding to the null and alternative hypotheses.

In the H_0 condition, two independent trajectories $y^{(1)}$ and $y^{(2)}$ are simulated from the same underlying dynamical system, i.e.,

$$(A^{(1)}, C^{(1)}, L^{(1)}) = (A^{(2)}, C^{(2)}, L^{(2)}).$$

A $\text{VAR}(p)$ model is then fitted to each trajectory, yielding estimated autoregressive coefficients $\{\hat{\Phi}_i^{(1)}\}_{i=1}^p$ and $\{\hat{\Phi}_i^{(2)}\}_{i=1}^p$. The distance between the resulting representations is then computed as described in Section 3.4.

In the H_1 condition, two independent trajectories $y^{(3)}$ and $y^{(4)}$ are simulated from two distinct underlying dynamical systems, i.e.,

$$(A^{(3)}, C^{(3)}, L^{(3)}) \neq (A^{(4)}, C^{(4)}, L^{(4)}).$$

Their corresponding $\text{VAR}(p)$ parameter estimates are then compared in the same manner.

Consequently, the distribution of the distance under H_0 reflects estimation variability within a single system, while the distribution under H_1 captures differences between independently sampled systems. The degree of separation between these two distributions determines the effectiveness of the proposed comparison method.

4.5 Distribution and Thresholding

For each realization, repeated Monte Carlo trials produce empirical distributions of the distance under both H_0 and H_1 . These distributions are estimated using kernel density estimation and summarized through descriptive statistics such as the mean, standard deviation, and upper quantiles.

The threshold τ used for classification is selected based on these empirical distributions, allowing us to quantify the trade-off between false positives and false negatives in practice. Performance metrics are then computed across all trials to evaluate the effectiveness of the proposed method.

4.6 Parameter Sweeps

To assess the robustness of the proposed comparison method, we perform a series of parameter sweeps over key quantities governing model complexity, statistical variability, and dynamical behavior.

These parameters are grouped as follows:

- **Model Complexity**

- **VAR order p** : controls the dimensionality of the autoregressive representation and determines the extent to which the underlying dynamics are captured.
- **Latent dimension d_x** : determines the complexity of the underlying dynamical system.
- **Observation dimension d_y** : controls the dimensionality of the observed process and, consequently, the size of the estimated parameter space.

- **Statistical Variability**

- **Effective sample size** $T = n - p$: governs the amount of data available for parameter estimation, directly impacting estimation accuracy.
- **Noise scale** σ : controls the magnitude of the Gaussian noise driving the system, allowing us to evaluate robustness under varying signal-to-noise conditions.

- **Dynamical Regime**

- **Regime separation** δ : controls the gap between the spectral radius intervals used to generate distinct systems in the H_1 condition. Specifically, the regimes are defined as

$$R_1 = [a, a + w], \quad R_2 = [a + w + \delta, a + 2w + \delta],$$

so that δ determines the separation between the two intervals.

For each parameter, we repeat the full experimental procedure described above while varying the parameter of interest and holding all others fixed at their baseline values. This allows us to isolate the effect of each parameter on classification performance.

Performance is evaluated using the metrics described in Section 3.5, with particular emphasis on AUC and F1 score. For each configuration, the optimal threshold τ is selected based on the empirical distributions of the distance under H_0 and H_1 .

These sweeps provide insight into how model complexity, statistical variability, and dynamical properties influence the separability of the induced parameter representations, and therefore the effectiveness of the proposed comparison method.

Because these parameter sweeps require repeated Monte Carlo evaluation across multiple parameter configurations, realizations, and trials, they were implemented using parallel configuration. A brief description of this implementation is provided in Appendix 4

5. Results

5.1 Distance Distributions

We begin by examining the empirical distributions of the isotropic distance under both the null and alternative hypotheses. These distributions are obtained by pooling results across multiple realizations and trials.

For this initial analysis, we use a fixed baseline configuration with parameters chosen without tuning:

$$p = 10, \quad d_x = 5, \quad d_y = 5, \quad n = 500,$$

and a baseline dynamical regime for the H_0 condition given by

$$\rho(F) \in (0.75, 0.85),$$

so that both trajectories are generated from systems within the same spectral radius interval.

For the H_1 condition, we consider distinct dynamical regimes defined by

$$R_1 \in [0.70, 0.85], \quad R_2 \in [0.85, 0.95],$$

corresponding to two non-overlapping spectral radius intervals for $F = A - LC$.

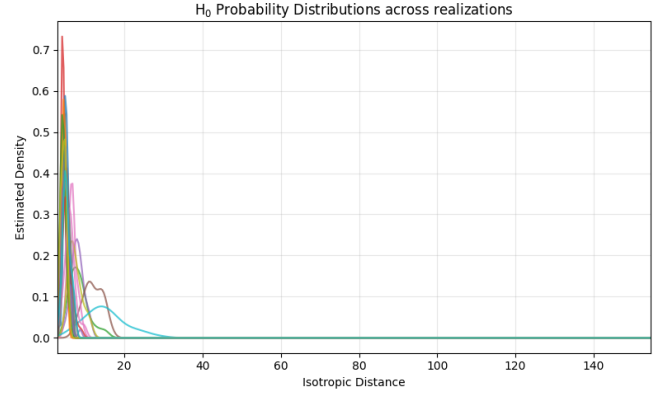


Figure 2. Empirical distributions of the isotropic distance under H_0 across realizations. Note that the distributions are centered near zero.

Under H_0 , the distance distributions seen in Figure 2 are highly concentrated near zero, reflecting the fact that independent trajectories generated from the same system yield similar autoregressive representations.

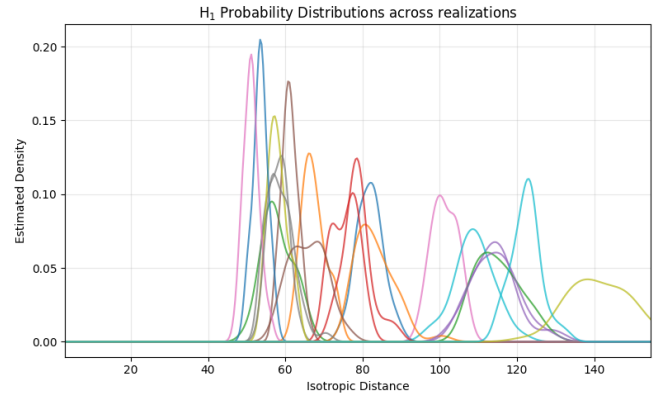


Figure 3. Empirical distributions of the isotropic distance under H_1 across realizations. Note the heterogeneity in both the location and spread of the distributions across realizations.

In contrast, under H_1 , the distance distributions seen in Figure 3 are shifted away from zero and exhibit significantly greater variability, reflecting structural differences between independently sampled systems.

To illustrate this separation more clearly, we consider a representative realization. As shown in Figure 4, the distributions under H_0 and H_1 are well-separated, enabling accurate classification via a simple thresholding rule. However, the optimal threshold τ , selected based on the F1 score, does not yield perfect separation. In particular, a small portion of the H_0 distribution lies to the right of the threshold, resulting in type I errors (false positives). This reflects the fact that, with a limited amount of data, the estimated autoregressive parameters are still subject to variability.

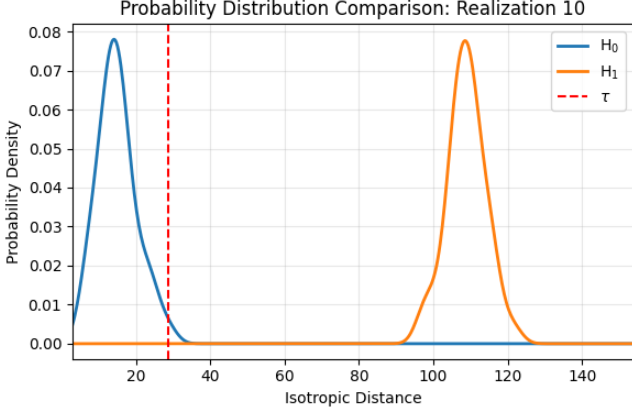


Figure 4. Comparison of H_0 and H_1 distance distributions for a single realization, with decision threshold τ . Note how the right tail of the H_0 distribution is to the right of the threshold, which yields Type I error(s).

5.2 Performance as a Function of Effective Sample Size

Having established that the isotropic distance provides clear separation between systems under a representative baseline configuration, we now examine how this separation depends on the amount of available data.

In particular, we vary the effective sample size $T = n - p$ while holding all other parameters fixed, and evaluate classification performance using AUC, F1 score, and the optimal decision threshold τ .

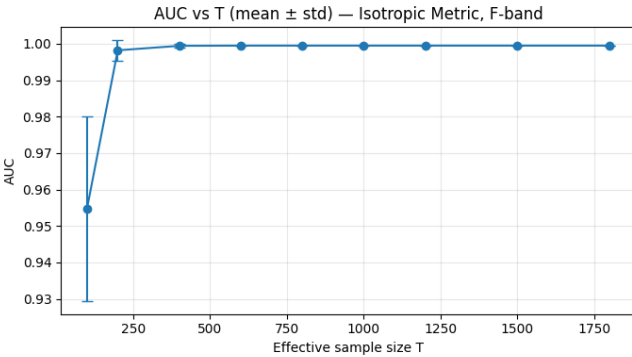


Figure 5. AUC as a function of effective sample size T (mean $\pm$ standard deviation across 100 repeats). Note the saturation beyond an effective sample size of $T > 250$.

As shown in Figure 5, the AUC increases rapidly with T , approaching near-perfect classification performance even for moderate sample sizes. This indicates that the autoregressive representations become increasingly stable and separable as more data is available.

In contrast, Figure 6 shows that the optimal threshold τ decreases as T increases, eventually stabilizing for larger sample sizes. This behavior is consistent with the reduction in estimation variability: as the variance of the estimated autoregressive parameters decreases, the distance between trajectories generated from the same system concentrates more tightly near zero, allowing for a lower decision threshold.

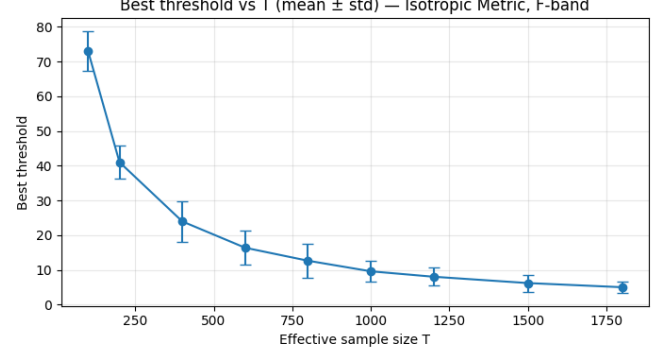


Figure 6. Optimal threshold τ as a function of effective sample size T (mean $\pm$ standard deviation across 100 repeats). The behavior as the effective sample size T increases is that of a monotonic decrease.

Taken together, these results demonstrate that classification performance saturates rapidly as the effective sample size increases. In particular, beyond a moderate value of T , the problem becomes nearly trivial, with near-perfect separability between the H_0 and H_1 distributions.

Consequently, in the parameter sweeps that follow, we adopt a relatively small sample size in order to avoid saturation effects and preserve sensitivity to variations in model complexity and dynamical regime. This ensures that differences in performance across parameter settings remain informative rather than being obscured by an abundance of data.

5.3 Parameter Sweep Analysis

We now examine how classification performance varies as a function of key model and data parameters. Across all experiments, performance is governed by a common trade-off between two competing effects:

- **Dynamical separability:** the extent to which distinct systems produce different autoregressive representations.
- **Estimation variability:** the difficulty of accurately estimating these representations from finite data.

We organize the results into three categories: model complexity, dynamical regime structure, and observational noise.

5.3.1 Model Complexity: p , d_y , and d_x

We first examine parameters that control the dimensionality of the autoregressive representation, namely the VAR order p , the observation dimension d_y , and the latent dimension d_x .

As shown in Figures 7 and 8, increasing p and d_y leads to a gradual degradation in classification performance. In both cases, the dimensionality of the parameter space grows, increasing estimation variability and reducing the separation between the H_0 and H_1 distributions. This effect is accompanied by an increase in the optimal threshold τ , reflecting the larger scale of distances in higher-dimensional spaces.

In contrast, the dependence on the latent dimension d_x is non-monotonic (Figure 9). Performance improves initially as d_x increases, reflecting richer underlying dynamics and greater

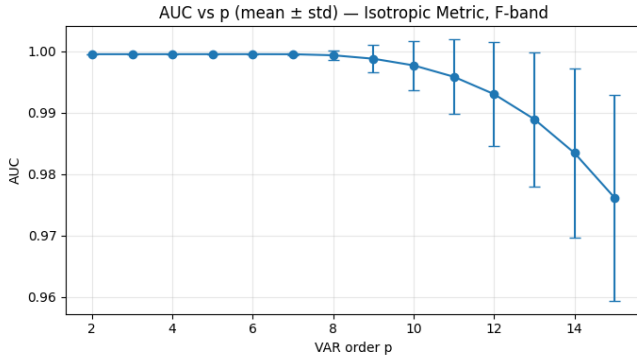


Figure 7. AUC as a function of VAR order p (mean $\pm$ standard deviation across 100 repeats) Note decrease in AUC and increase in variance as the VAR order grows.

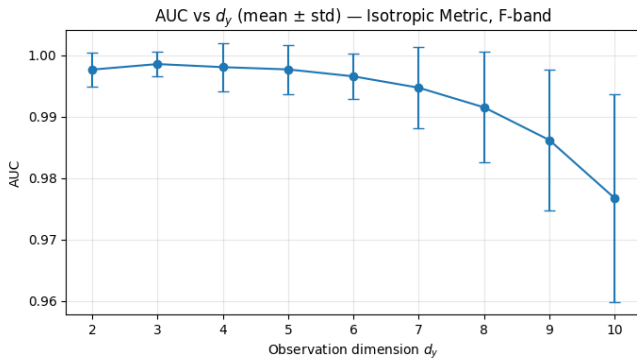


Figure 8. AUC as a function of observation dimension d_y (mean $\pm$ standard deviation across 100 repeats). The behavior of d_y is similar to that of p , insofar that as the dimension grows, the variance grows and the AUC decreases.

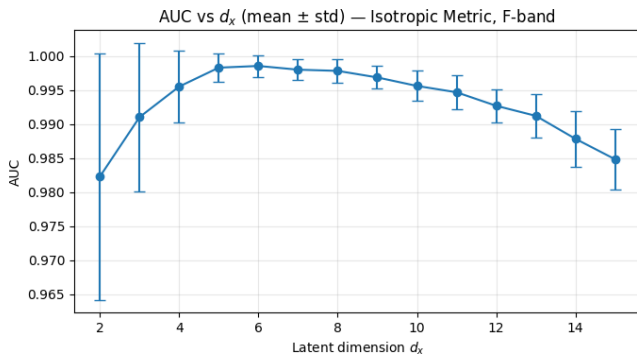


Figure 9. AUC as a function of latent dimension d_x (mean $\pm$ standard deviation across 100 repeats). Note the inverse behavior of d_x as it pertains to d_y , i.e., the variance is greater for lower dimensions.

diversity between independently sampled systems. However, beyond moderate values of d_x , performance begins to decline, suggesting that increased latent complexity introduces additional estimation variability.

When considered jointly, these results highlight a fundamental trade-off: increasing model complexity enhances representational capacity but can degrade performance under finite-sample conditions.

5.3.2 Dynamical Regime Separation

We next examine the effect of dynamical separation between systems under H_1 . This is controlled by constructing two non-overlapping spectral radius intervals for the effective memory operator $F = A - LC$:

$$R_1 = [a, a + w], \quad R_2 = [a + w + \delta, a + 2w + \delta],$$

where δ determines the gap between the two regimes.

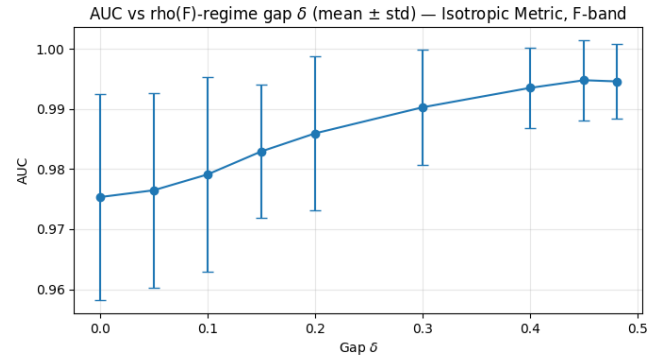


Figure 10. AUC as a function of regime separation δ (mean $\pm$ standard deviation across 100 repeats).

As shown in Figure 10, classification performance improves monotonically with δ . Larger values of δ correspond to greater differences in the spectral properties of F , which in turn produce more distinct autoregressive representations. As a result, the distributions of the distance under H_0 and H_1 become increasingly well-separated.

This result provides direct empirical support for the role of F in governing observable dynamics: separation in the spectral radius of F translates into separation in the induced autoregressive parameter space.

5.3.3 Robustness to Observation Noise

Finally, we examine the effect of observation noise by varying the scale σ of the Gaussian disturbance $\epsilon_k \sim \mathcal{N}(0, \sigma^2 I)$.

As shown in Figure 11, classification performance remains essentially constant across a wide range of noise levels. This indicates that the proposed method is highly robust to observational perturbations.

This robustness arises because the comparison is performed in the space of estimated autoregressive parameters. While noise affects individual trajectories, it impacts both H_0 and H_1

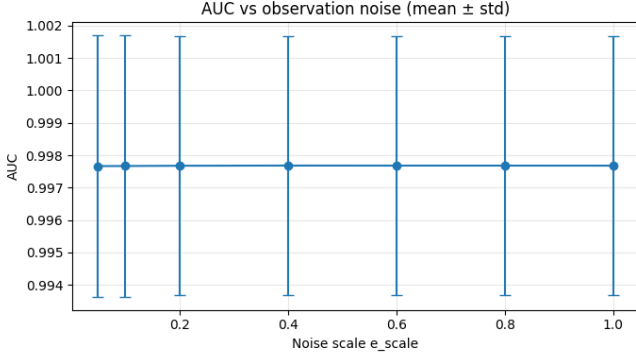


Figure 11. AUC as a function of observation noise scale σ (mean $\pm$ standard deviation across 100 repeats). Note the lack of change in both mean and variance, signaling that our model is robust to noise.

distributions in a similar manner, preserving their relative separation. Consequently, the method captures intrinsic dynamical structure rather than being sensitive to measurement-level noise.

Overall, these results demonstrate that the proposed framework performs best when the underlying systems are sufficiently distinct and the model complexity is appropriately matched to the available data, while remaining robust to observation noise.

5.4 Experimental Analysis Under Optimal Conditions

Having examined the influence of individual parameters, we now consider the performance of the proposed method under a configuration informed by the preceding analysis. In particular, we select parameters that balance representational capacity and statistical stability, namely moderate values of p , d_x , and d_y , together with a sufficiently large sample size n to ensure accurate estimation of the autoregressive coefficients.

Using this configuration, we recompute the empirical distributions of the isotropic distance under both H_0 and H_1 , pooling results across realizations and trials.

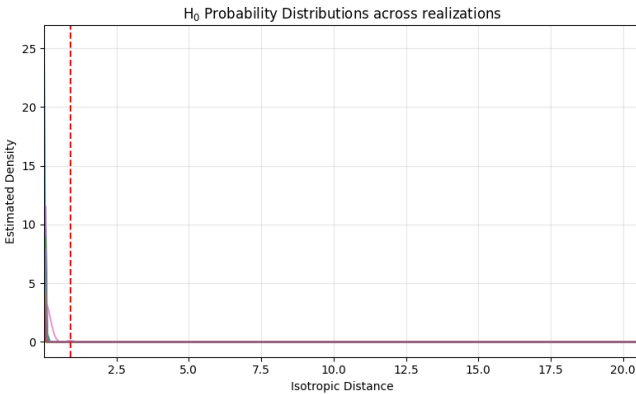


Figure 12. Pooled distributions of the isotropic distance under H_0 across realizations using the optimized parameter configuration. Note the higher concentration toward zero and dramatic decrease in magnitude relative to the naïve H_0 distributions in 2

As shown in Figure 12, the distribution under H_0 remains more tightly concentrated near zero than in the baseline configuration, reflecting the consistency of the estimated autoregressive representations when trajectories are generated from the same underlying system. The narrow spread indicates reduced estimation variability under the chosen parameter regime.

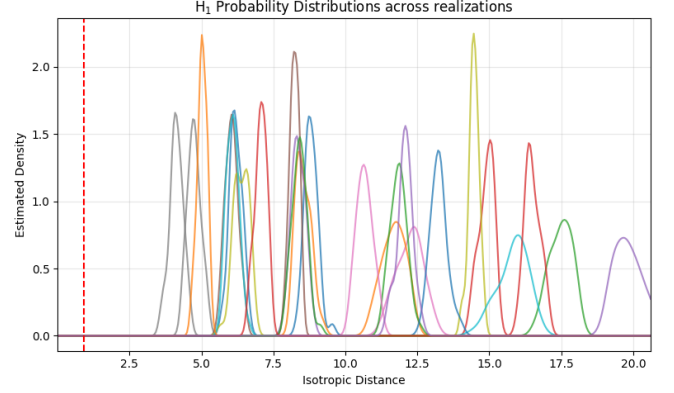


Figure 13. Pooled distributions of the isotropic distance under H_1 across realizations using the optimized parameter configuration. Note the decrease in magnitude relative to the naïve H_1 distributions in 3

In contrast, Figure 13 shows that the distribution under H_1 is well-separated from that of H_0 , with distances concentrated at larger values. This separation reflects the increased distinguishability of systems with differing dynamical regimes when represented through their autoregressive coefficients.

Compared to the baseline configuration introduced in Section 5.1, the optimized setting yields a more pronounced gap between the two distributions. In particular, the overlap between H_0 and H_1 is further reduced, leading to near-perfect separability using a simple thresholding rule.

These results demonstrate that, when appropriately tuned, the proposed framework provides a highly effective means of distinguishing between linear dynamical systems based solely on observed trajectories. Moreover, the strong separation observed here confirms the central premise of the method: that differences in latent dynamics are faithfully captured in the induced autoregressive representations, and can be robustly detected using an isotropic distance metric.

5.5 Hypothesis Testing Performance Under Optimal Conditions

We now quantify classification performance under the optimized parameter configuration described in the previous subsection. Using the threshold selected via F1-score maximization, we evaluate the resulting hypothesis test on pooled samples across realizations.

The resulting performance metrics are summarized as follows:

$$\text{AUC} = 0.9995, \quad \tau^* = 0.9112,$$

$$\text{Accuracy} = 1.0000, \quad \text{Precision} = 1.0000, \quad \text{Recall} = 1.0000,$$

$$\text{FPR} = 0.0000, \quad \text{FNR} = 0.0000, \quad \text{F1} = 1.0000.$$

Under this configuration, the classifier achieves perfect separation between the H_0 and H_1 classes on the sampled dataset, with zero false positives and zero false negatives. In particular, all trajectories generated from the same system are correctly identified as such, while all trajectories generated from distinct systems are correctly distinguished.

This result is consistent with the strong separation observed in the empirical distance distributions (Figures 12–13). The absence of overlap between the two distributions allows for a threshold τ^* that perfectly partitions the sample space.

While this level of performance reflects a favorable configuration (combining sufficient sample size, moderate model complexity, and clear separation in dynamical regimes) it provides a useful upper bound on the effectiveness of the proposed method. In practice, performance will depend on the interplay between these factors, as explored in the preceding parameter sweeps.

Nevertheless, these results demonstrate that, when estimation error is sufficiently controlled and dynamical differences are pronounced, the isotropic distance between autoregressive representations provides a highly reliable criterion for distinguishing between linear dynamical systems.

6. Discussion

The results of this work highlight a consistent relationship between latent dynamical structure and its observable representation. In particular, differences in the underlying system (captured by the effective memory operator $F = A - LC$) are reliably reflected in the estimated autoregressive coefficients. This provides a practical means of comparing dynamical systems without requiring explicit recovery of latent parameters.

A central theme emerging from the experiments is the trade-off between representational capacity and statistical reliability. Increasing model complexity, through larger values of p , d_x , or d_y , enhances the expressiveness of the autoregressive representation but also increases estimation variability in finite-sample settings. Conversely, increasing the amount of data stabilizes the estimated parameters, leading to rapid improvements in classification performance and, in many cases, near-perfect separation between systems.

The experiments also demonstrate that the proposed method is sensitive to differences in dynamical regimes. As the separation between spectral radius intervals increases, the corresponding autoregressive representations become more distinct, resulting in improved classification performance. This reinforces the role of F as the key object governing the observable behavior of the system.

Another notable finding is the robustness of the method to observation noise. Across a wide range of noise levels, classification performance remains largely unchanged. This suggests that the comparison operates at the level of underlying structure rather than being driven by variability in individual observations.

Despite these strengths, several limitations remain. The present study relies on synthetic data generated under Gaussian assumptions and considers only autonomous systems without

external inputs. In addition, the decision threshold τ is selected empirically from the observed distance distributions, rather than derived analytically. A particularly promising direction for future work is the development of a closed-form or theoretically grounded approximation for τ as a function of model parameters and available data. Such a result would provide a principled decision rule independent of simulation, and would clarify how estimation variability, model complexity, and dynamical separation jointly determine classification performance. Extending the framework to more general settings, including real-world data and alternative noise structures, also remains an important avenue for further investigation.

7. Conclusion

In summation, we introduced a framework for comparing linear dynamical systems using only observed time series. By approximating each observed process through a finite-order VAR model and comparing the resulting autoregressive representations with an isotropic distance, we reduce the problem of system comparison to a hypothesis test in an observable parameter space.

Through systematic Monte Carlo experiments, we showed that the proposed method can reliably distinguish between systems across a range of settings. Performance improves with sample size, degrades gracefully as model complexity increases, improves with greater dynamical regime separation, and remains notably robust to observation noise. Under a tuned parameter configuration, the method achieves near-perfect discrimination between the null and alternative hypotheses.

These results demonstrate that latent dynamical differences can be effectively detected without explicit recovery of the underlying state-space parameters. This provides a simple and practical alternative to direct system identification, while also suggesting a clear path toward future theoretical work, particularly in deriving principled thresholding rules for system comparison.

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Appendix 1. Derivation of the VAR Representation

Starting from the LDS:

$$x_{k+1} = Ax_k + L\epsilon_k$$

$$y_k = Cx_k + \epsilon_k$$

we recursively expand the state:

$$x_k = (A - LC)x_{k-1} + Ly_{k-1}.$$

Iterating this relation and under the stability condition $\rho(A - LC) < 1$, the resulting series converges and yields:

$$x_k = \sum_{i=1}^{\infty} (A - LC)^{i-1} L y_{k-i}.$$

Substituting into the observation equation:

$$y_k = Cx_k + \epsilon_k,$$

we obtain:

$$y_k = \sum_{i=1}^{\infty} C(A - LC)^{i-1} L y_{k-i} + u_k.$$

Thus, the coefficients are given by:

$$\Phi_i = C(A - LC)^{i-1} L.$$

Appendix 2. Least Squares Derivation

We begin by examining the estimator $\hat{B}$, which is defined as the unique matrix that attains the global minimum of the quadratic objective. Formally:

$$\hat{B} = \arg \min_B \|Y - XB\|_F^2$$

where,

X : Design matrix constructed from past observations

Y : Response matrix containing current observations

B : Unknown coefficient matrix to be estimated

XB : Model prediction of Y

We then define the objective function:

$$f(B) = \|Y - XB\|_F^2$$

which expands via the Frobenius norm identity to,

$$\|Y\|_F^2 + \|XB\|_F^2 - 2\langle Y, XB \rangle_F.$$

Using linearity of the Gradient operator and standard identities from matrix calculus (i.e., the derivative of a quadratic Frobenius norm and the derivative of a Frobenius inner product respectively), we differentiate each term individually:

$$\nabla_B f(B) = \underbrace{\nabla_B \|Y\|_F^2}_{(i.)} + \underbrace{\nabla_B \|XB\|_F^2}_{(ii.)} - 2 \underbrace{\nabla_B \langle Y, XB \rangle_F}_{(iii.)}$$

$$(i.) \nabla_B \|Y\|_F^2 = 0$$

$$(ii.) \nabla_B \|XB\|_F^2 = 2X^\top XB$$

$$(iii.) \nabla_B \langle Y, XB \rangle_F = X^\top Y$$

Then we simply set the gradient to zero and solve for B :

$$\Rightarrow 0 = 2X^\top XB - 2X^\top Y$$

$$\Rightarrow X^\top XB = X^\top Y$$

$$\Rightarrow \hat{B} = (X^\top X)^{-1} (X^\top Y)$$

Appendix 3. Sampling Normalization Across Dimension Sweeps

In the baseline experiments, the observation matrix C is sampled from a Gaussian distribution and the noise injection matrix L is taken to be an identity-type map. In particular, when d_x and d_y differ, $L \in \mathbb{R}^{d_x \times d_y}$ is chosen as the rectangular identity truncation, so that noise enters the system without introducing additional directional structure.

When sweeping model dimensions, however, naive Gaussian sampling can introduce changes in scale that are not directly attributable to the parameter being varied. In particular, when varying the latent dimension d_x , sampling

$$C_{\text{raw}} \sim \mathcal{N}(0, 1)^{d_y \times d_x}$$

without normalization would cause the Frobenius norm of C_{raw} to grow systematically with dimension. As a consequence, the magnitude of the induced terms LC and $C(A - LC)^{i-1}L$ could change simply because the matrix dimension has changed, rather than because of any meaningful dynamical effect.

To reduce this confounding effect, we normalize the sampled observation matrix according to

$$C = \frac{C_{\text{raw}}}{\|C_{\text{raw}}\|_F}.$$

This keeps the overall scale of C comparable across different values of d_x , so that changes in classification performance are more directly attributable to changes in latent dimension rather than unintended scaling effects.

More generally, similar normalization or rescaling procedures could be applied to L when non-identity choices are used. The purpose of these adjustments is not to alter the qualitative structure of the experiment, but rather to ensure that parameter sweeps remain controlled and interpretable across different system configurations.

Appendix 4. Parallelization of Parameter Sweeps

The parameter sweeps conducted in this work require repeated Monte Carlo simulations across many realizations and trials, leading to substantial computational cost. To mitigate this, we employed parallel computation using the `joblib` library in Python.

Parallelization is applied at the level of independent sweeps (e.g., over d_y or repeated trials), allowing each configuration to be evaluated concurrently across available CPU cores. Care is taken to preserve reproducibility by controlling random seeds

within each parallel worker, which ensures that results remain consistent with a serial implementation.

Crucially, this approach enables efficient exploration of the parameter space without altering the statistical structure of each experiment.

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